DO NOW — RECALL

# Client-side versus server-side processing

Name: _____ Class: _____ Date: _____

**1. What does an HTML form submit?**

**2. What does JavaScript textContent do?**

**3. What does GET generally do with form data?**

**4. A 2D array has 3 rows and 4 columns. Give total cells and last zero-based indexes.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Name/value fields to a server using a chosen HTTP method.

### 2. Model answer

Changes text in a DOM node without parsing it as HTML.

### 3. Model answer

Encodes it as query parameters in the request URL.

### 4. Model answer

12 cells; last index pair (2,3).

### Vocabulary — know these terms

client-side — Processing on user's browser/device.

server-side — Processing on service infrastructure.

validation — Checking data against rules.

authentication — Verifying user identity.

authorisation — Deciding permitted actions.

NEW LEARNING

# Read, annotate and explain

### Understand

Client-side JavaScript can offer immediate feedback and responsive interfaces, but its validation can be bypassed or altered.

### Apply

Server-side code enforces permissions, validates input against business rules and updates authoritative records or databases.

### Evaluate

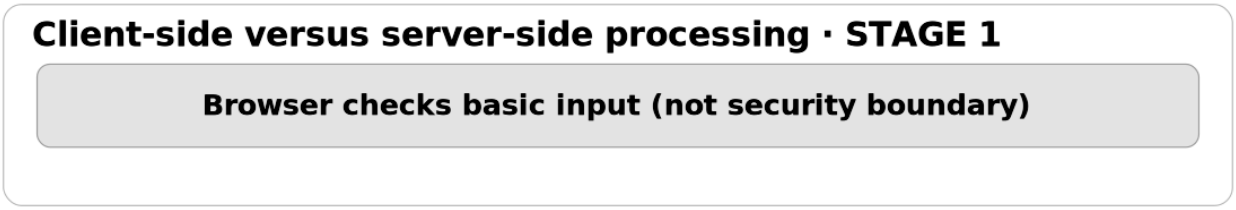
Processing choices trade off device load, server load, confidentiality, latency, accessibility and reliability; many systems use both.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

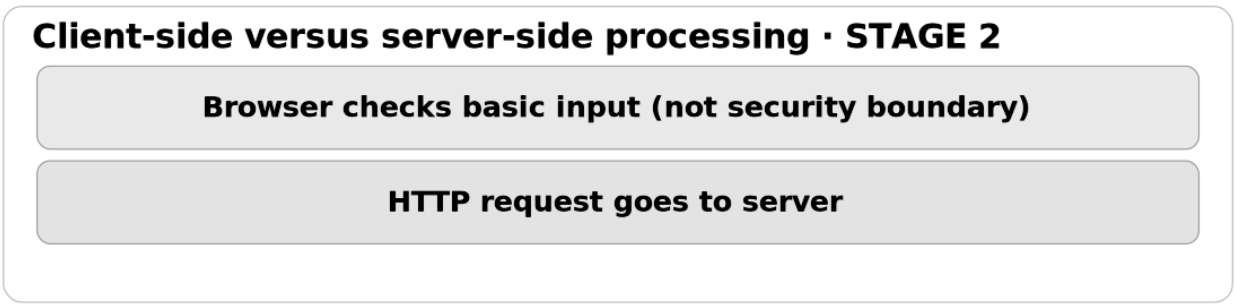
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning



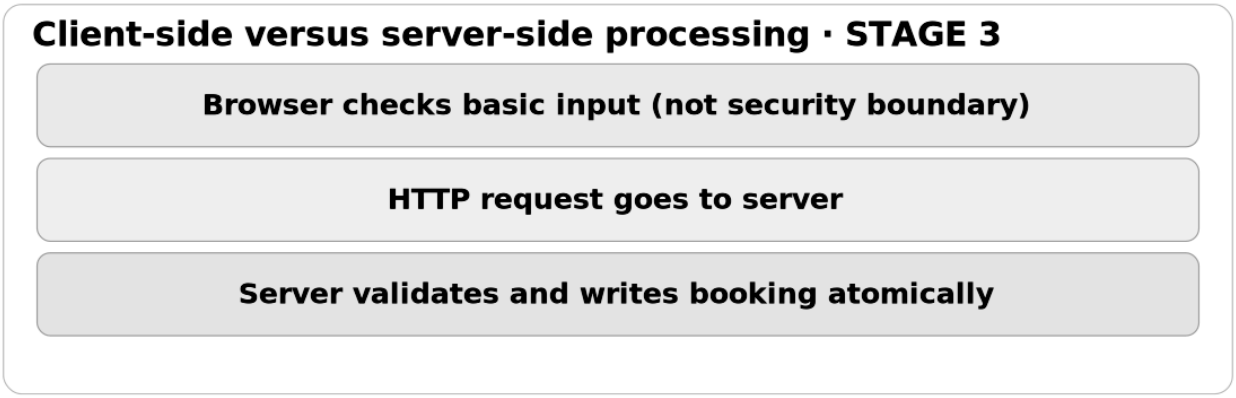

Client-side versus server-side processing · STAGE 1; Browser checks basic input (not security boundary)

### New Learning




Client-side versus server-side processing · STAGE 2; Browser checks basic input (not security boundary) then HTTP request goes to server

### New Learning

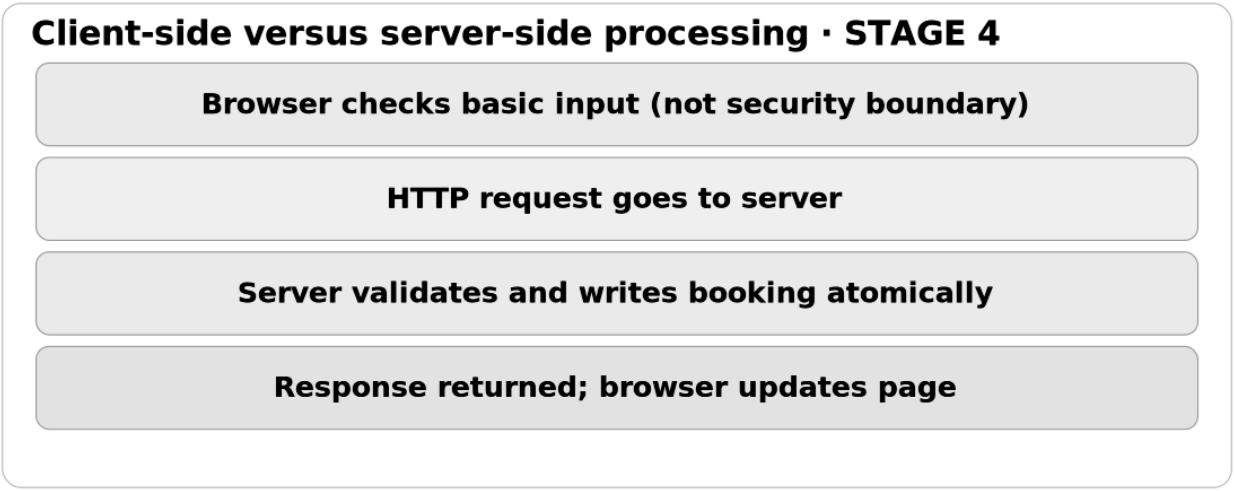



Client-side versus server-side processing · STAGE 3; Browser checks basic input (not security boundary) then HTTP request goes to server then Server validates and writes booking atomically

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Client-side versus server-side processing · STAGE 4; Browser checks basic input (not security boundary) then HTTP request goes to server then Server validates and writes booking atomically then Response returned; browser updates page

I DO — WORKED EXAMPLE

# Follow the worked method

### Why must a server check a submitted quantity even if the browser requires 1–5?

### Predict first

An attacker can skip the browser form and send a request containing quantity=1000.

### Explain

The server must enforce actual constraints; browser checks are useful but only improve convenience.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Why must a server check a submitted quantity even if the browser requires 1–5?

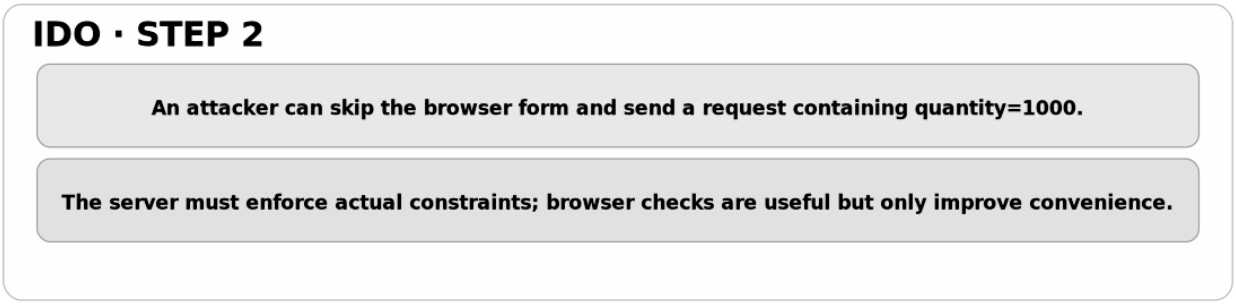
An attacker can skip the browser form and send a request containing quantity=1000.




IDO · STEP 1; An attacker can skip the browser form and send a request containing quantity=1000.

### Why must a server check a submitted quantity even if the browser requires 1–5?

An attacker can skip the browser form and send a request containing quantity=1000.




IDO · STEP 2; An attacker can skip the browser form and send a request containing quantity=1000. then The server must enforce actual constraints; browser checks are useful but only improve convenience.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A pupil updates their profile photo. Decide which processing belongs in browser versus server.

The browser can preview or resize an image to reduce traffic and improve usability.




WEDO · STEP 1; The browser can preview or resize an image to reduce traffic and improve usability.

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first The browser can preview or resize an image to reduce traffic and improve usability.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Compare client-side and server-side processing using form validation. [6]

### Question 2

An online shop must deduct stock and charge for an order. Explain which operations cannot rely solely on a browser. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Evaluate a hybrid design for an accessible online examination practice system. [6]

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A server can trust form data solely because the browser used required.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A pupil updates their profile photo. Decide which processing belongs in browser versus server.

The browser can preview or resize an image to reduce traffic and improve usability.




WEDO · STEP 2; The browser can preview or resize an image to reduce traffic and improve usability. then The server verifies authorisation, validates the uploaded bytes and applies safe storage rules.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Explain

The server verifies authorisation, validates the uploaded bytes and applies safe storage rules.

IMPROVE: Explain one correction to your own reasoning.